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GLIDE: A Python library for prediction-powered evaluation of GenAI systems

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Software

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Summary

Deciding whether a generative AI (GenAI) system behaves acceptably is a measurement problem: what fraction of answers contain hallucinations, or what fraction of an agent's trajectories are unsafe? Human expert annotation answers such questions reliably but slowly and at high cost, while the popular LLM-as-judge paradigm is cheap but yields biased proxy annotations (Zheng et al., 2023). Prediction-powered inference (PPI) dissolves that trade-off: it combines a small human-labeled sample (taken as ground truth) with a large pool of proxy-labeled observations into a debiased metric, whose confidence interval remains valid however poor the proxy is (Angelopoulos, Bates, et al., 2023; Angelopoulos, Duchi, et al., 2023).

`GLIDE` (Generated Label Inference & Debiasing Engine) brings that literature into a unified library, specialized in risk and performance estimation of complex AI systems that produce large amounts of unstructured data, such as AI agents. Most such metrics, be they accuracy, hallucination rate or unsafe-trajectory rate, are population means estimated on samples of generated outputs. Since human annotation is scarce, their estimation borrows from survey theory, where a small sample estimates a population statistic: it requires both choosing which observations humans should label and combining the resulting ground-truth labels with proxy labels.

`GLIDE` addresses both needs and exposes multiple sampling designs and performance estimators behind a scikit-learn-style API. Every estimator returns a result object carrying the point estimate, the confidence interval, and the effective sample size, i.e., the number of human labels a classical survey estimator would have needed to reach the same precision. Whenever the LLM-as-judge correlates with human annotators, the effective sample size translates into direct savings in human annotations.

`GLIDE` also addresses monitoring, which consists of repeated evaluations on a stream of batches, each carrying a few human labels and many proxy labels. Naively repeating an evaluation inflates the false alarm rate, so monitoring requires dedicated statistical control of multiple testing. `GLIDE` monitors return the running metric together with an anytime-valid bound (upper for a risk metric, lower for a performance metric).

The library is documented through runnable tutorials, a mathematical user guide, a scientific validation notebook for every estimator and monitor, and several end-to-end case studies on public benchmarks.

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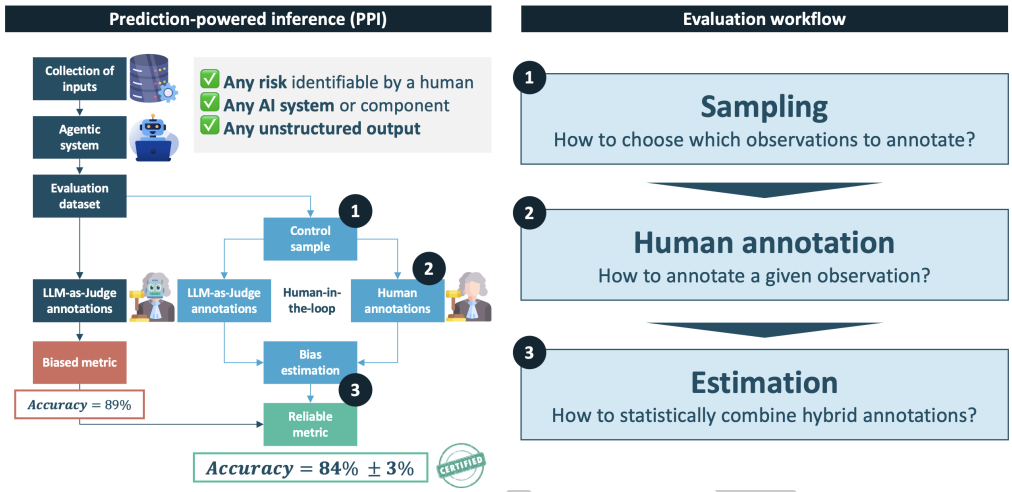


Figure 1: The evaluation workflow `GLIDE` implements. Left: prediction-powered inference debiases a cheap proxy metric using a small human-annotated control sample. Right: the resulting three-step decomposition around which the API is organized.

Statement of need

Evaluating a deployed GenAI system under a fixed annotation budget raises several complications, each addressed by a separate strand of the literature.

- **Heterogeneous strata.** LLM-as-judge quality varies across axes known in advance: domain, language, query type. Stratified sampling tunes evaluation per stratum (Fisch et al., 2024; Fogliato et al., 2024).
- **Small labeled sets.** Below a few dozen labels per stratum, the central-limit approximation degrades. Bootstrap intervals stay valid (Kluger et al., 2025).
- **Correlated units.** For granular metrics (e.g., the proportion of hallucinated sentences), evaluation units may not be independent (think of sentences within a paragraph). Treating them as independent understates the variance, so clustered inference, which works on group means, is needed.
- **Multiple judges.** Modern evaluation stacks accumulate LLM judges of differing cost and skill. Multi-proxy aggregation weights them optimally, so an extra judge can only narrow the interval (Cowen-Breen et al., 2026; Shan et al., 2025).
- **Judge uncertainty.** LLM judges can expose per-observation uncertainty scores that reveal where a human label pays off. Active sampling concentrates the budget there (Gligorić et al., 2025; Zrnic & Candes, 2024).
- **Cost asymmetry.** Expert review costs dollars per unit, an LLM-as-judge call cents. Cost-aware sampling targets human annotations at the most promising observations under a monetary budget (Angelopoulos et al., 2025).
- **Continuous monitoring.** Re-estimating a metric batch after batch inevitably produces false alarms. Anytime-valid confidence sequences make peeking safe under a single budget (Podkopaev & Ramdas, 2021; Waudby-Smith et al., 2024; Zhang et al., 2026).

These methods share one template: a few reliable annotations combined with a large set of proxy annotations. However, they and their extensions are scattered across papers with heterogeneous notation and partial reference implementations (Song et al., 2026), some in R and some in Python, none covering more than a slice of the design space.

`GLIDE` closes that gap by treating sampling, estimation and monitoring as one consistent family.

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State of the field

ppi_py (Angelopoulos, Bates, et al., 2023) has long been the reference implementation of the PPI family. This open-source Python package covers means, generalized linear models and M-estimators. The remaining methods above live in single-paper repositories: ssepy for stratified sampling and estimation (Fogliato et al., 2024), active-inference and confidence-driven-inference for active designs (Gligorić et al., 2025; Zrnic & Candes, 2024), PTDBoot for the bootstrap variants (Kluger et al., 2025), and sada for multi-proxy aggregation (Shan et al., 2025). Each repository is self-consistent, but has its own data conventions, no shared sampling layer and no monitoring counterpart.

Evaluation orchestration frameworks such as RAGAS (Es et al., 2024), DeepEval (Ip & Vongthongsri, 2026), TruLens (TruEra, 2026) and Inspect (UK AI Security Institute, 2024) solve the engineering problem of running an LLM-as-judge evaluation. GLIDE can thus consume the outputs of such pipelines, combine them with ground-truth labels, and supply the downstream rigorous statistical estimation layer.

We built a new library rather than contributing these methods upstream for two reasons.

The first is audience and scope. ppi_py is the reference implementation of the seminal PPI work and remains the right tool for inference on general estimands (e.g., quantiles, regression coefficients). It starts from an already-labeled set, so sampling design falls outside its scope. GLIDE addresses the broader community of engineers and evaluation teams, and buys that accessibility by narrowing the estimand to the mean, the workhorse of system evaluation. The narrowing also buys breadth: ppi_py predates most of the extensions listed above, whereas GLIDE brings cost-aware and active designs, stratified and clustered variants, multi-proxy aggregation, bootstrap intervals and anytime-valid monitoring into one consistent framework, several of them in their first public implementation.

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The second reason follows from the first. Serving that audience requires an API that reconciles sampling, estimation and monitoring in one modular architecture. GLIDE provides a unified API that lets users switch easily between methods, and lets contributors add a new estimator, sampler or monitor without having to understand or modify the rest of the code base.

Software design

GLIDE organizes evaluation around three sequential steps inherited from survey theory (Figure 1): sampling, which selects which observations deserve a human label; annotation, which only domain experts can perform; and estimation, which combines the labels and the proxy predictions into a debiased estimate with a confidence interval. Monitoring is a fourth component that wraps per-batch estimates in an anytime-valid bound.

Two design decisions carry most of the library's behavior.

The first is the data contract: each class method take as inputs one aligned array per collection of labels, following the conventions scikit-learn established across the data science community (Pedregosa et al., 2011).

The second is a middle layer of abstraction: each estimator family factors its computation into a MeanEstimationEngine, which an estimator calls once over the whole dataset and a monitor

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calls once per batch. Because estimators and monitors share this middle layer, contributors can readily extend any estimator into its anytime-valid monitor counterpart, without reimplementing its statistics.

In terms of quality gates, every new component passes three lines of defense, all run in continuous integration: unit tests with 100% code coverage and executed docstring examples, functional tests of statistical properties (e.g., a stratified estimator on a single stratum must equal the uniform one), and Monte Carlo validation notebooks assessing statistical coverage, interval width, false-alarm and miscoverage rates.

Regarding maintenance philosophy, `scipy` (Virtanen et al., 2020) is the single runtime dependency, our dependency support windows follow SPEC 0, and releases follow semantic versioning.

Research impact statement

The statistical framework behind the library is described in a companion methods paper presented at the ICML 2026 workshop on statistical frameworks for uncertainty in agentic systems (Martinon et al., 2026). Three reproducible case studies in the documentation run full workflows on public benchmarks, whose proxy-labeled versions we release as documented datasets: agentic safety evaluation on R-Judge (Yuan et al., 2024), text-to-SQL accuracy on Spider (Yu et al., 2018), and multilingual retrieval-augmented generation faithfulness on MEMERAG (Cruz Blandón et al., 2025).

`GLIDE` has reached practitioner and research audiences alike. It was the subject of a tutorial at PyData Amsterdam 2026 (Martinon, 2026a), will be presented at Compute! Paris 2026 (Martinon, 2026b) and at the AI4Good workshop at NeurIPS 2026 in Paris (Merad & Martinon, 2026).

`GLIDE` has been developed in public since its first commit in March 2026, with roughly two releases per month and a dozen contributors. It has attracted more than a hundred stars and is disseminated through a monthly newsletter. Beyond the open-source community, the package has already been used in several missions delivered by Emerton Data, a consulting firm, to evaluate industrialized GenAI systems with limited human annotation budgets.

Code availability

The package can be installed from PyPI with `pip install glide-py`. The source is on GitHub (https://github.com/EmertonData/glide).

AI usage disclosure

Generative AI was used both to develop the software and to prepare this manuscript. Development tasks are co-designed by the tech lead with Claude Code. Developers co-implement with Claude Code. Each pull request is co-reviewed: Claude's code-review first, then a tech lead's review. For the manuscript, Claude was used to brainstorm, format, and audit the draft against the journal's guidelines. The authors take full responsibility of the result.

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